

TRT PROTOCOLS

BLOOD WORK GUIDE

HORMONE OPTIMIZATION

SIDE EFFECT MANAGEMENT

TESTOSTERONE REPLACEMENT THERAPY (TRT) COMPLETE OPTIMIZATION GUIDE

Protocols · Dosing · Blood Work · Aromatase Management · Thyroid · Sexual Health

Evidence-based hormone optimization for health, vitality & body composition

ELITE FITNESS GUIDE SERIES

Research-Based · Professional Edition · 2024

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1. LOW TESTOSTERONE — SYMPTOMS, CAUSES & DIAGNOSIS

Low testosterone (hypogonadism) affects an estimated **2–4% of adult men**, with subclinical low-normal testosterone (300–450 ng/dL) affecting a much larger population. Symptoms often present years before lab values fall below clinical cutoffs, causing significant quality-of-life impairment.

■ A major American urological study found that men with total testosterone <300 ng/dL have a 42% increased cardiovascular risk, significantly reduced bone density, and a 3× higher incidence of metabolic syndrome compared to men with optimal T levels (600–900 ng/dL).

Symptoms of Low Testosterone:

Category	Symptoms	Severity
Physical	Decreased muscle mass, increased body fat (especially abdominal), reduced strength, fatigue even after adequate sleep, decreased body fat, frequently missed for years	
Sexual	Reduced libido, erectile dysfunction, decreased morning erections, reduced ejaculate volume, infertility	Major quality of life impact; often first symptom noticed
Psychological	Depression, anxiety, irritability, reduced motivation, cognitive fog, poor concentration	Often misdiagnosed as depression — antidepressants are NOT the solution if cause is testosterone
Metabolic	Insulin resistance, type 2 diabetes risk, dyslipidemia (high LDL, low HDL, triglycerides)	Forces a negative feedback loop — fat cells convert T to estrogen, worsening T deficiency

2. THE HYPOTHALAMIC-PITUITARY-TESTICULAR (HPT) AXIS

Understanding the HPT axis is essential for interpreting blood work, understanding why TRT suppresses natural production, and making informed decisions about HCG, AI, and PCT:

- **Hypothalamus** releases GnRH (Gonadotropin-Releasing Hormone) in pulses every 1–3 hours
- **Pituitary gland** responds to GnRH by releasing LH (Luteinizing Hormone) and FSH (Follicle-Stimulating Hormone)
- **LH** travels to Leydig cells in the testes → stimulates testosterone production
- **FSH** travels to Sertoli cells → stimulates sperm production (spermatogenesis)
- **Testosterone** feeds back to hypothalamus and pituitary → reduces GnRH, LH, FSH (negative feedback)
- **TRT effect:** Exogenous testosterone triggers strong negative feedback → LH and FSH fall to near zero → testes stop producing testosterone and sperm → testicular atrophy over months

This is why **HCG is critical** on TRT for men who want to maintain fertility or prevent complete testicular atrophy.

3. TRT PROTOCOLS — TYPES, DOSES & ADMINISTRATION

Protocol	Dose	Frequency	Pros	Cons
Testosterone Enanthate (IM)	100–200 mg	Every 7–14 days	Inexpensive; familiar; stable	Peaks and troughs with 14-day schedule; requires injection
Testosterone Enanthate (Sub-Q)	80–150 mg	Every 5–7 days (split)	More stable levels; less SHBG elevation; less SHBG elevation	Requires more frequent injections
Testosterone Cypionate (IM)	100–200 mg	Every 7–14 days	Slightly longer half-life than Enanthate; widely available	Same as Enanthate
Testosterone Propionate	25–50 mg	Every 2–3 days	Very stable blood levels; fast adjustment	Daily or EOD injections; more painful
Testosterone Gel/Cream	50–100 mg topical	Daily application	Non-invasive; very stable levels	Transfer risk to women/children; variable absorption
Testosterone Pellets	Implanted pellets	Every 3–6 months	Zero administration burden	Surgical insertion; dose adjustment difficult

Optimal TRT Dose Principle: The goal is to bring total testosterone into the **upper-normal physiological range (700–1000 ng/dL)**, not simply above the clinical cutoff (300 ng/dL). Many men feel optimal at 900+ ng/dL while still remaining within a physiological range that minimizes cardiovascular and prostate risk.

4. FREE VS TOTAL TESTOSTERONE — OPTIMIZATION

Total testosterone measures ALL testosterone in blood. However, only **2–4% of testosterone is 'free'** (unbound to proteins) and biologically active. The rest is bound to SHBG (Sex Hormone Binding Globulin) or albumin.

■ Research demonstrates that free testosterone is a superior predictor of clinical outcomes (muscle mass, libido, energy) compared to total testosterone. A man with total T of 700 ng/dL but very high SHBG may have the same symptoms as a man with total T of 400 ng/dL and normal SHBG.

How to Optimize Free Testosterone:

- **Reduce SHBG:** SHBG rises with age, insulin sensitivity, thyroid dysfunction, and low androgen levels. TRT itself reduces SHBG somewhat

- **Optimize thyroid:** Hypothyroidism dramatically increases SHBG — correcting T3/T4 liberates bound testosterone
- **Maintain healthy body weight:** Obesity increases SHBG and aromatase activity simultaneously
- **Adequate zinc:** Zinc inhibits SHBG synthesis in the liver — supplement 25–50 mg/day
- **Boron:** 10 mg/day boron has been shown to reduce SHBG by 9% and increase free T by 25% in a clinical study
- **Injection frequency:** More frequent TRT injections (2–3×/week) maintain more stable free T levels vs once-weekly peaks

■ 5. ESTROGEN MANAGEMENT ON TRT

Estrogen management on TRT requires a more nuanced approach than during cycle use. **Men on TRT need adequate estrogen** for cardiovascular health, bone density, cognitive function, and sexual desire. Aggressive AI use on TRT is one of the most common and harmful mistakes.

■■ **IMPORTANT:** Multiple studies show that men with very low estrogen on TRT have **WORSE** cardiovascular outcomes than men with optimally elevated estrogen. Do **NOT** attempt to suppress E2 to low levels — it increases atherosclerosis risk, reduces bone density, and causes sexual dysfunction.

TRT Estrogen Targets:

- Total T 700–1000 ng/dL with E2 30–50 pg/mL = optimal for most men (some feel best at E2 40–60 pg/mL)
- Symptoms matter more than numbers — adjust AI based on how you feel AND blood work together
- If AI is needed: Anastrozole 0.25 mg twice weekly maximum — start lower than AAS protocols
- Consider Aromasin over Arimidex for TRT — suicidal AI provides less fluctuation in E2 levels

■ 6. HCG ON TRT — FERTILITY & TESTICULAR HEALTH

- **Standard TRT dose:** 500 IU HCG 2× weekly alongside TRT injections
- **Fertility-focused:** 1000–2000 IU 3× weekly + consider FSH (Gonal-F or Follistim) if sperm count remains low
- **Testicular atrophy prevention:** 250 IU EOD (smaller, more frequent) maintains testicular size better than large infrequent doses
- **Mechanism:** HCG mimics LH — directly stimulates Leydig cells in testes to maintain function and size
- **HCG storage:** Reconstitute with bacteriostatic water; refrigerate; stable 60 days

■ 7–9. THYROID · DHT · BLOOD WORK GUIDE

Thyroid Optimization on TRT:

The thyroid-testosterone relationship is bidirectional — low T reduces T3/T4 conversion, and low thyroid further depresses testosterone. Testing TSH alone is insufficient; demand **full thyroid panel** (TSH, Free T3, Free T4, Reverse T3, TPO antibodies).

Test	Optimal Range	Action if Out of Range
TSH	0.5–2.0 mIU/L	TSH >2.5 with symptoms: consider thyroid support
Free T3	3.5–4.2 pg/mL	Low Free T3 with normal T4: impaired T4→T3 conversion; consider T3 supplementation
Free T4	1.2–1.7 ng/dL	Low T4: primary hypothyroidism; may require T4 (Synthroid) or combination T4+T3
Reverse T3	<15 ng/dL	High rT3 = T3 is being blocked; often caused by chronic stress or caloric restriction

DHT & Hair Loss Management:

- Testosterone converts to DHT via 5-alpha reductase enzyme — DHT is the primary cause of male pattern baldness (MPB) in genetically susceptible men
- 5-alpha reductase inhibitors (Finasteride 1mg, Dutasteride 0.5mg) reduce scalp DHT by 60–90%
- Caution: Finasteride can cause permanent sexual side effects ('Post-Finasteride Syndrome') in a small percentage of men — research thoroughly before use
- Alternative: Topical Finasteride (0.025–0.1% solution applied to scalp) — reduces scalp DHT with minimal systemic absorption

Complete TRT Blood Work Panel:

Frequency	Panel
Baseline (before TRT)	Total T, Free T, LH, FSH, E2, Prolactin, SHBG, DHT, CBC, CMP, Lipids, PSA, TSH, Free T3, Free T4, HbA1c, Ferritin, Vitamin D
3 months post-start	Total T, Free T, E2, SHBG, CBC (hematocrit), PSA, Lipids, CMP
Every 6 months (stable)	Total T, Free T, E2, SHBG, CBC, PSA, Lipids, CMP
Annually	Full panel same as baseline

♥ 10–12. CARDIOVASCULAR · SEXUAL HEALTH · LIFESTYLE

Cardiovascular Monitoring on TRT:

- TRT increases red blood cell production — check hematocrit every 3–6 months. If >52%: donate blood or reduce dose
- TRT may worsen dyslipidemia (especially HDL). Omega-3 4–6g/day is non-negotiable
- Blood pressure: Monitor monthly; target <130/80. Reduce sodium; exercise cardiovascular training
- Recent large studies (TRAVERSE trial, 2023) found TRT does NOT increase cardiovascular events in hypogonadal men with pre-existing cardiovascular disease

Optimizing Sexual Function on TRT:

- Libido usually improves within 3–6 weeks of TRT initiation
- If libido remains low despite optimal T: check E2 (too high or too low), prolactin (elevated = dopamine issue), thyroid (low T3), and zinc/vitamin D deficiency
- PT-141 (Bremelanotide): 1–2 mg SC 45–90 min before activity — acts centrally on desire circuits regardless of hormonal status
- Tadalafil (Cialis): 5 mg daily (low-dose) improves blood flow and may improve T response by increasing testicular blood circulation

Lifestyle Synergy with TRT: Resistance training 4×/week (amplifies TRT response by upregulating AR density). Sleep 7–9 hours (testosterone peaks during REM sleep — poor sleep negates TRT). Maintain body fat 10–18% (higher fat = more aromatization; lower fat = less conversion). Minimize alcohol (directly suppresses Leydig cell function).